



MAHIR LABIB CHOWDHURY

Mechanical Engineer, CSWA

☎ 01747088882

✉ mahirlc12@gmail.com

📍 Adabar, Dhaka-1207

🌐 www.linkedin.com/in/mahir-labib-chowdhury

🌐 <https://mahirlc12.wixsite.com/mahirlc> (Portfolio Website)

EXPERIENCE

Mechanical Design Engineer Intern

Flymek Aerospace, Dhaka | Oct 2025 – Dec 2025

- Designed and developed mechanical components and assemblies using SolidWorks
- Prepared detailed 3D models, manufacturing drawings with GD&T annotations
- Performed engineering calculations for structural integrity and material selection to meet design requirements
- Supported prototyping, testing, and validation cycles, iterating designs based on test data
- Applied DFM principles to optimize part weight, durability, and production cost

Freelance Mechanical Design Engineer

Independent (Fiverr, Upwork) | Dec 2025 – Present

- Designing custom mechanical components and assemblies based on client requirements
- Delivering CAD models and technical drawings

PROJECT & LEADERSHIP EXPERIENCE

Team Leader & Structure Designer – Robotics Team (Team Thunderbolt)

- Led a team of 5 in the design, development, and fabrication of competition robots for national events
- Designed the mechanical structure with focus on strength, weight optimization, and ease of assembly
- Managed project execution, task allocation, and timeline coordination
- Coordinated across mechanical, electrical, and software workstreams during integration

Selected Engineering Projects

- Designed and fabricated a vertical axis wind turbine prototype, performing aerodynamic analysis and engineering calculations for blade design and structural integrity
- Developed and optimized airfoil design for a light aircraft using XFLR5 and parametric simulation, achieving measurable performance improvement in lift-to-drag ratio (6.9%)
- Designed and simulated belt drive mechanisms, performing power transmission calculations and validating performance through simulation
- Designed a flywheel energy storage system for AC backup during power outages, performing structural and thermal analysis in ANSYS and dynamic modeling in MATLAB/Simulink

EDUCATION

Bachelor of Science (BSc) in Mechanical Engineering | CGPA: 3.62

2025

Ahsanullah University of Science and Technology (AUST), Dhaka

Higher Secondary Certificate (HSC) | GPA: 5.00

2019

Birshreshtha Munshi Abdur Rouf Public College, Dhaka

SUMMARY

Mechanical Engineer with hands-on experience in mechanical design and product development. Proficient in SolidWorks (CSWA), ANSYS (CFD/FEA), and engineering documentation including 3D models, manufacturing drawings with GD&T, and BOMs. Proven ability to develop concepts into functional prototypes through iterative design, testing, and cross-functional coordination with teams. Strong foundation in engineering calculations, material selection, and design for manufacturing (DFM). Passionate about innovative product development and cost-optimized mechanical solutions.

CORE SKILLS

CAD & Visualization

- SolidWorks
- AutoCAD
- Fusion 360
- KeyShot

Mechanical Design & Development

- Product Development & Rapid Prototyping
- Design for Manufacturing (DFM)
- GD&T
- Engineering Calculations
- Material Selection
- BOM Preparation

Simulation & Engineering Analysis

- ANSYS Fluent (CFD), ANSYS Mechanical (FEA- Structural, Thermal)
- MATLAB / Simulink
- Aerodynamic Analysis (XFLR5)

Manufacturing Process Knowledge

- Machining (CNC Milling, Turning)
- Sheet Metal Fabrication
- 3D Printing / Additive Manufacturing
- Welding & Assembly Processes

Engineering Fundamentals

- Technical Documentation & Reporting
- Feasibility Analysis & Cost Optimization
- Microsoft Office (Word, Excel, PowerPoint)

CERTIFICATIONS

Certified SOLIDWORKS Associate (CSWA)

Score: 235/240

Certified Supply Chain Analyst

Score: 91/100